



DW_rbsh

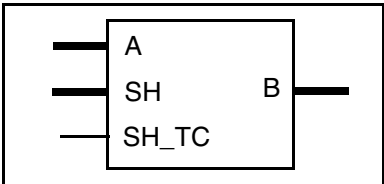
Barrel Shifter with Preferred Right Direction

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Features and Benefits

- Parameterized data and shift coefficient word lengths
- Capable of rotating in both directions
- Inferable using a function call

Revision History



Description

DW_rbsh is a general-purpose barrel shifter that prefers to rotate the information in the right direction. The input data A is rotated right or left by the number of bit positions specified by the control input SH. Rotation implies that bits shifted out wrap around from the LSB to the MSB when it is a right shift, and from the MSB to the LSB when it is a left shift. The recommended maximum value of the parameter *SH_width* is related to *A_width* by the equation:

$$SH_width = \text{ceil}(\log_2[A_width])$$

Smaller values of *SH_width* can be used to save hardware. *SH_width* can be larger than $\text{ceil}(\log_2[A_width])$, but this choice usually generates unnecessary hardware.

Table 1-1 Pin Description

Pin Name	Width	Direction	Function
A	<i>A_width</i>	Input	Input data
SH	<i>SH_width</i>	Input	Shift control
SH_TC	1 bit	Input	Shift two's complement control ■ 0 = Unsigned ■ 1 = Signed
B	<i>A_width</i>	Output	Shifted data out

Table 1-2 Parameter Description

Parameter	Values	Description
<i>A_width</i>	≥ 1 (see Table 1-4 on page 2)	Word length of A and B
<i>SH_width</i>	$\leq \text{ceil}(\log_2[A_width])$	Word length of SH

Table 1-3 Synthesis Implementations^a

Implementation Name	Function	License Feature Required
str	Synthesis model target for speed	DesignWare
astr	Synthesis model target for area	DesignWare
mx2	Implement using 2:1 multiplexers only The mx2 implementation is valid only for <i>SH_width</i> values up to and including 31.	none

- a. During synthesis, Synopsys synthesis tools select the appropriate architecture for your constraints. Alternatively, you may use the `set_implementation` command to choose one specific implementation from those listed in this table. For details, see the documentation for your Synopsys synthesis tool.

The following table lists some values for *A_width* and the corresponding *SH_width* values. For example, if *A_width* = 8, *SH_width* = 3 is adequate because there can be at most $2^3 = 8$ shift combinations.

When *SH_TC* = 0, the coefficient *SH* is interpreted as an unsigned positive number and DW_rbsh performs only right rotate operations.

When *SH_TC* = 1, *SH* is interpreted as a two's complement number. A negative *SH* value performs a left rotate and a positive *SH* value performs a right rotate.

Table 1-4 Sample Parameter Values

A_width	SH_width
1 - 2	1
3 - 4	2
5 - 8	3
9 - 16	4
17 - 32	5
33 - 64	6

Table 1-5 Simulation Models

Model	Function
dw/dw01/src/DW_rbsh_sim.vhd	VHDL simulation model source code
dw/sim_ver/DW_rbsh.v	Verilog simulation model source code

Table 1-6 Truth Table ($A_width = 8$, $SH_width = 3$)

SH(2:0)	SH_TC	B(7)	B(6)	B(5)	B(4)	B(3)	B(2)	B(1)	B(0)
000	X	A(7)	A(6)	A(5)	A(4)	A(3)	A(2)	A(1)	A(0)
001	X	A(0)	A(7)	A(6)	A(5)	A(4)	A(3)	A(2)	A(1)
010	X	A(1)	A(0)	A(7)	A(6)	A(5)	A(4)	A(3)	A(2)
011	X	A(2)	A(1)	A(0)	A(7)	A(6)	A(5)	A(4)	A(3)
100	0	A(3)	A(2)	A(1)	A(0)	A(7)	A(6)	A(5)	A(4)
101	0	A(4)	A(3)	A(2)	A(1)	A(0)	A(7)	A(6)	A(5)
110	0	A(5)	A(4)	A(3)	A(2)	A(1)	A(0)	A(7)	A(6)
111	0	A(6)	A(5)	A(4)	A(3)	A(2)	A(1)	A(0)	A(7)
100	1	A(3)	A(2)	A(1)	A(0)	A(7)	A(6)	A(5)	A(4)
101	1	A(4)	A(3)	A(2)	A(1)	A(0)	A(7)	A(6)	A(5)
110	1	A(5)	A(4)	A(3)	A(2)	A(1)	A(0)	A(7)	A(6)
111	1	A(6)	A(5)	A(4)	A(3)	A(2)	A(1)	A(0)	A(7)

Related Topics

- [Logic – Combinational Overview](#)
- [DesignWare Building Block IP User Guide](#)

HDL Usage Through Function Inferencing - VHDL

```
library IEEE,DWARE;
use IEEE.std_logic_1164.all;
use IEEE.std_logic_arith.all;
use DWARE.DWpackages.all;
use DWARE.DW_foundation_arith.all;

entity DW_rbsh_func is
    generic (
        func_A_width : POSITIVE := 8;
        func_SH_width : POSITIVE := 3
    );
    port (
        func_A : in std_logic_vector(func_A_width-1 downto 0);
        func_SH : in std_logic_vector(func_SH_width-1 downto 0);
        func_SH_TC : in std_logic;
        B_func : out std_logic_vector(func_A_width-1 downto 0)
    );
end DW_rbsh_func;

architecture func of DW_rbsh_func is

begin

    -- Functional inference of DW_rbsh
    B_func <= std_logic_vector(DWF_rbsh(UNSIGNED(func_A),UNSIGNED(func_SH)) );

end func;
```

HDL Usage Through Function Inferencing - Verilog

```
module DW_rbsh_func( func_A, func_SH, func_SH_TC, B_func );

parameter func_A_width = 8;
parameter func_SH_width = 3;

// secondary parameters used to pass parameters to function
// when parameter names differ

parameter A_width = func_A_width;
parameter SH_width = func_SH_width;

// Please add search_path = search_path + {synopsys_root + "/dw/sim_ver"}
// to your .synopsys_dc.setup file (for synthesis) and add
// +incdir+$SYNOPSYS/dw/sim_ver+ to your verilog simulator command line
// (for simulation).
`include "DW_rbsh_function.inc"

input [func_A_width-1 : 0] func_A;
input [func_SH_width-1 : 0] func_SH;
input func_SH_TC;
output [func_A_width-1 : 0] B_func;

// Infer DW_rbsh

assign B_func = DWF_rbsh_uns(func_A, func_SH);

endmodule
```

HDL Usage Through Component Instantiation - VHDL

```
library IEEE,DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DWpackages.all;
use DWARE.dw_foundation_comp.all;

entity DW_rbsh_inst is
    generic (
        inst_A_width : POSITIVE := 8;
        inst_SH_width : POSITIVE := 3
    );
    port (
        inst_A : in std_logic_vector(inst_A_width-1 downto 0);
        inst_SH : in std_logic_vector(inst_SH_width-1 downto 0);
        inst_SH_TC : in std_logic;
        B_inst : out std_logic_vector(inst_A_width-1 downto 0)
    );
end DW_rbsh_inst;

architecture inst of DW_rbsh_inst is

begin

    -- Instance of DW_rbsh
    U1 : DW_rbsh
    generic map (
        A_width => inst_A_width,
        SH_width => inst_SH_width
    )
    port map (
        A => inst_A,
        SH => inst_SH,
        SH_TC => inst_SH_TC,
        B => B_inst
    );

end inst;
```

HDL Usage Through Component Instantiation - Verilog

```
module DW_rbsh_inst( inst_A, inst_SH, inst_SH_TC, B_inst );

parameter A_width = 8;
parameter SH_width = 3;

input [A_width-1 : 0] inst_A;
input [SH_width-1 : 0] inst_SH;
input inst_SH_TC;
output [A_width-1 : 0] B_inst;

    // Instance of DW_rbsh
    DW_rbsh #(A_width, SH_width) U1 (
        .A(inst_A),
        .SH(inst_SH),
        .SH_TC(inst_SH_TC),
        .B(B_inst) );

endmodule
```

Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

Date	Release	Updates
September 2024	DWBB_202409.0	■ Updated date and version; no technical changes
December 2023	DWBB_202312.0	■ Updated date and version; no technical changes ■ Added this Revision History table and the document links on this page

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